

Injection units for dosing pumps

Installation and operating instructions



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GRUNDFOS 

English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe the Grundfos injection units for dosing pumps.

Sections 1-3 give the information necessary to be able to install the product in a safe way.

Sections 4-7 give important information about the product as well as information on service and disposal of the product.

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Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General information

1.1 Target group

This document is intended for the operating company and the users. It contains general instructions that must be observed before installation and during operation and maintenance of the product. The responsible staff must read these instructions prior to any work on the product.

1.1.1 Qualification and training

The persons responsible for the tasks described in this document must be appropriately qualified.

1.1.2 Obligations of the operating company

- Observe the local safety regulations.
- Keep the installation and operating instructions available at the installation location at all times.
- Coordinate the preparation of the installation location observing section 6. *Technical data*.
- Ensure that the users are trained for their tasks.
- Provide the stipulated safety equipment and personal protective equipment.
- Arrange regular maintenance.

1.1.3 Obligations of the user

- Observe the recognised health and safety regulations as well as the local accident prevention regulations.
- Wear protective equipment in accordance with local health and safety regulations when working on the product and handling chemicals.
- Read and understand this document.

1.2 Symbols used in this document



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.

1.3 Safe operation

When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.



Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Only operate the system if all lines are connected correctly.

If safe operation is no longer possible, the product must be taken out of operation and secured against unintentional operation.

This is the case in the following situations:

- If the product is visibly damaged.
- If the product does not seem operational.
- After long periods of storage under unfavourable conditions.

2. Installing the product

2.1 Location

- The installation location must be protected from rain, humidity, condensation, direct sunlight and dust.
- The installation location must have sufficient lighting to ensure safe operation.
- Observe the permissible ambient conditions. See section [6.2 Storage and ambient temperatures](#).

2.2 Mechanical installation

The product must only be installed by authorised and qualified persons.

When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.



Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Observe section [3.1 Handling the product](#).

Wipe up spilled liquid immediately to avoid slipping hazard.

2.2.1 Installation example

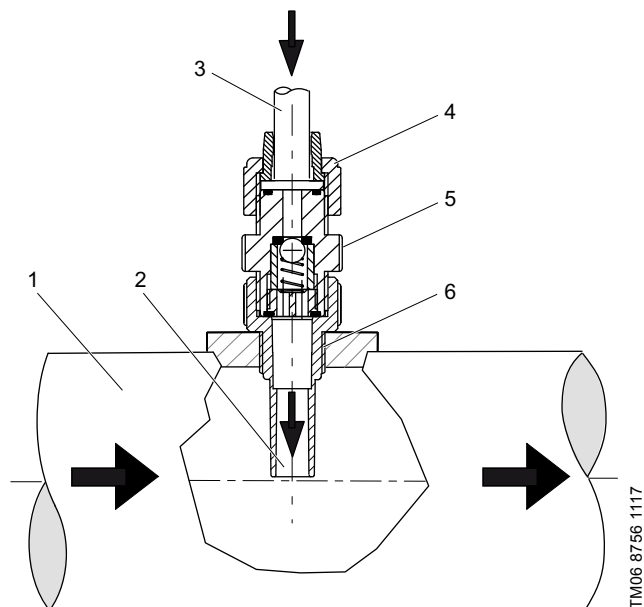


Fig. 1 Installation example

Pos.	Description
1	Process line
2	Injection pipe
3	Dosing line
4	Dosing line connection
5	Spring loaded non-return valve
6	Process line thread

2.2.2 Connections

Injection units up to 1500 l/h and high-pressure injection units have a process line thread for the process line connection and are supplied with a tube or pipe connection set for the dosing line connection.

Injection units up to 4000 l/h have flange connections on both sides.



Make sure that the non-return valve of the injection unit is installed in flow direction. The flow direction is indicated by an arrow on the valve body.

2.2.3 Installing standard injection units and injection units with ball valve up to 6, 60, 460 and 1500 l/h and high-pressure injection units

1. If necessary, shorten the injection pipe.
 - The injection pipe outlet should be as close as possible to the centre of the process line.
2. Apply appropriate sealing material to the thread of the process line connection.
3. Carefully screw the injection unit into the process line.
4. Connect the dosing line. Observe the separate manual of the dosing pump.

2.2.4 Installing standard injection units up to 4000 l/h



Injection units with flange connection must be installed at the bottom side of horizontal process lines. Otherwise the non-return valve may not work correctly.

1. If necessary, shorten the injection pipe.
 - The injection pipe outlet should be as close as possible to the centre of the process line.
2. Install the injection unit vertically from the bottom at the flange connection of the process line.
3. Connect the dosing line. Observe the separate manual of the dosing pump.

2.2.5 Installing injection units with minimal immersion depth or with lip valve

Observe sections:

- [4.2.2 Injection unit with minimal immersion depth](#)
- [4.2.3 Injection unit with lip valve](#)

The injection pipes of these injection units cannot be shortened. The injection pipe outlet should be as close as possible to the centre of the process line.

1. For injection units with lip valve:
 - Apply appropriate sealing material to the thread of the process line connection.
2. For injection units with minimal immersion depth:
 - Make sure that the flat gasket of the process line connection is positioned correctly or apply appropriate sealing material to the thread.
3. Carefully screw the injection unit into the process line.
4. Connect the dosing line. Observe the separate manual of the dosing pump.

2.2.6 Installing injection units with removable injection pipe

Observe section [4.2.5 Injection unit with removable injection pipe](#).



In normal operation, the injection pipe passes through the ball valve and the process line connection. Do not try to close the ball valve in normal operation, otherwise the injection pipe can be damaged.

Observe section [5.2 Removing and cleaning the removable injection pipe](#).

1. If necessary, shorten the injection pipe or adjust the immersion depth via the clamping ring connection. See fig. 7, pos. 3.
 - The injection pipe outlet should be as close as possible to the centre of the process line.
2. Apply appropriate sealing material to the thread of the process line connection.
3. Carefully screw the injection unit into the process line.
4. Connect the dosing line. Observe the separate manual of the dosing pump.

2.2.7 Installing hot-injection units with ball valve

Observe section [4.2.6 Hot-injection unit with ball valve](#).

The injection pipe outlet should be as close as possible to the centre of the process line.

1. Apply appropriate sealing material to the thread of the process line connection.
2. Carefully screw the injection unit into the process line.
3. If necessary, shorten the cooling pipe.



The cooling pipe must be long enough to ensure proper cooling function.

Deburr the pipe ends of the cooling pipe after shortening.

If bending the cooling pipe, make sure that the cross-section is not reduced.

4. Connect the cooling pipe to the ball valve.
 - Tighten the cutting ring connection thoroughly.
5. Connect the non-return valve to the cooling pipe.
 - Tighten the cutting ring connection thoroughly.
6. Connect the dosing line. Observe the separate manual of the dosing pump.

3. Handling and storing the product

3.1 Handling the product

- Wear safety shoes during transport and installation.
- Wear a helmet if lifting the product higher than hip level.
- Use appropriate lifting and transporting equipment.
- Make sure that the product is not exposed to any point load during the transport.
- Avoid strong impacts.
- Observe the permissible ambient conditions. See section [6.2 Storage and ambient temperatures](#).

3.2 Storing the product

- Observe the permissible ambient conditions. See section [6.2 Storage and ambient temperatures](#).
- The storage location must be protected from rain, humidity, condensation, direct sunlight and dust.

4. Product introduction

4.1 Intended use

Injection units for dosing pumps are used in conjunction with Grundfos dosing pumps. They are intended for injecting the dosing medium either into a process line or into a tank where it is mixed with the process water.

Injection units are suitable for liquid, non-abrasive, non-flammable and non-combustible media.

Observe the freezing and boiling points of the medium.

Make sure that parts in contact with the medium are resistant to the medium under operating conditions.



The product must not be used for any other purpose than the one mentioned above.

4.1.1 Improper operating methods

The operational safety of the product is only guaranteed, if it is used in accordance with section [4.1 Intended use](#).



Injection units are not isolating valves. They only close completely if the system pressure falls below the opening pressure.

The product must not be used for:

- operation in potentially explosive areas
- frozen media or gases
- crystallising media.

4.2 Product variants

4.2.1 Standard injection unit

Standard injection units maintain a minimum counterpressure to avoid backflow of the dosing medium.

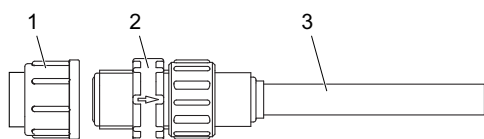


Fig. 2 Standard injection unit (example)

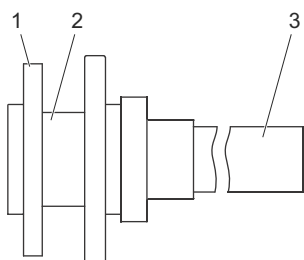


Fig. 3 Standard injection unit up to 4000 l/h

Pos.	Description
1	Dosing line connection set
2	Spring-loaded non-return valve
3	Injection pipe

For technical data see section:

- [6.3 Standard injection units up to 6 l/h](#)
- [6.5 Standard injection units up to 60 l/h](#)
- [6.10 Standard injection units up to 460 l/h](#)
- [6.12 Injection units up to 1500 l/h](#)
- [6.13 Injection units up to 4000 l/h](#)

4.2.2 Injection unit with minimal immersion depth

Injection units with minimal immersion depth are typically used for small process line diameters. An O-ring provides the non-return function.

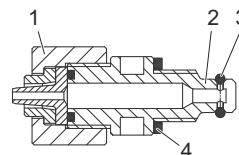


Fig. 4 Injection unit with minimal immersion depth

Pos.	Description
1	Dosing line connection set
2*	Injection pipe
3	O-ring for non-return function
4	Flat gasket

* The injection pipe cannot be shortened.

For technical data see section:

- [6.4 Injection units with minimal immersion depth up to 6 l/h](#)

4.2.3 Injection unit with lip valve

Injection units with lip valve are typically used to add sodium hypochlorite solution to water with a high carbonate content. The lip prevents crystallisation and blocking caused by alkali carbonate reactions at the point of injection.

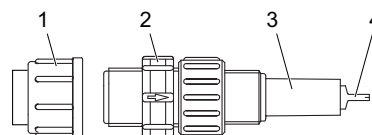


Fig. 5 Injection unit with lip valve

Pos.	Description
1	Dosing line connection set
2	Spring-loaded non-return valve
3*	Injection pipe
4	Lip valve

* The injection pipe of injection units with lip valve cannot be shortened.

For technical data see section:

- [6.6 Injection units with lip valve up to 60 l/h](#)

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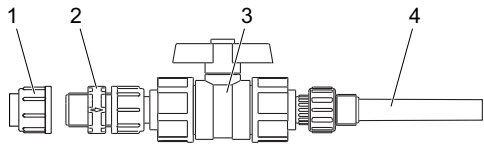
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4.2.4 Injection unit with ball valve

Injection units with ball valve are used for applications where the injection point must be closable. The ball valve is placed between the injection pipe and the spring-loaded non-return valve. Thus, the dosing line can be completely disconnected from the process. The non-return valve can be disassembled and cleaned without stopping the process and emptying the process line.



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Fig. 6 Injection unit with ball valve

Pos.	Description
1	Dosing line connection set
2	Spring-loaded non-return valve
3	Ball valve
4	Injection pipe

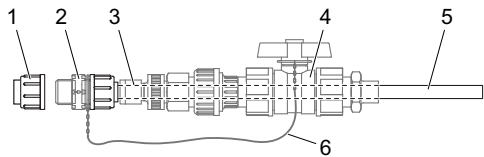
For technical data see section:

- 6.7 Injection units with ball valve up to 60 l/h
- 6.11 Injection units with ball valve up to 460 l/h

4.2.5 Injection unit with removable injection pipe

Injection units with removable injection pipe are used where regular cleaning of the injection pipe is required. The injection pipe can be removed from the process line without stopping the process water flow. The injection point can be closed with the integrated ball valve once the injection pipe was removed.

Observe section 5.2 Removing and cleaning the removable injection pipe.



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Fig. 7 Injection unit with removable injection pipe

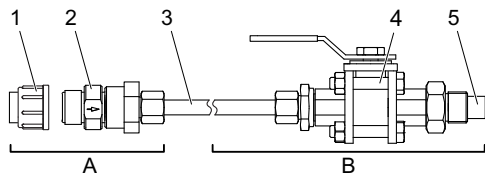
Pos.	Description
1	Dosing line connection set
2	Spring-loaded non-return valve
3	Clamping ring connection
4	Ball valve
5	Injection pipe
6	Safety cord

For technical data see section:

- 6.8 Injection units with removable injection pipe up to 60 l/h

4.2.6 Hot-injection unit with ball valve

Hot-injection units with ball valve can be used for direct injection of the dosing medium into processes with a high process water temperature. The cooling pipe (3) separates the hot parts that are installed in the process line from the non-return valve and the dosing line connection that have standard operating temperature.



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Fig. 8 Hot-injection unit with ball valve

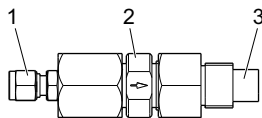
Pos.	Description
A	Area with standard operating temperature
B	Area with high operating temperature
1	Dosing line connection set
2	Spring-loaded non-return valve
3	Bendable cooling pipe
4	Ball valve
5	Injection pipe

For technical data see section:

- 6.9 Hot-injection units with ball valve up to 60 l/h

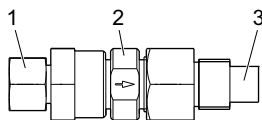
4.2.7 High-pressure injection units

High-pressure injection units are especially designed for high-pressure applications.



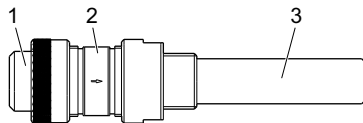
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Fig. 9 High-pressure injection unit, flow rate 3.3 l/h



TM07 0554 0218

Fig. 10 High-pressure injection unit, flow rate 55 l/h



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Fig. 11 High-pressure injection unit, flow rate 220 l/h

Pos.	Description
1	Dosing line connection set
2	Spring-loaded non-return valve
3	Injection pipe

For technical data see section:

- 6.14 High-pressure injection units

4.3 Identification

4.3.1 Type key

The type key is designed for the precise identification of the product and not for configuration purposes. It can be found on the product packaging.

Example: **INJ.UNIT 0300-10 PVC/E/C 4U3-30/120**

Product type	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
Injection unit	
Size	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
01	Up to 6 l/h
02	Up to 60 l/h
03	Up to 460 l/h
04	Up to 1500 l/h
05	Up to 4000 l/h
Variant	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
00	Standard
01	Process line connection material: 2.4610 (Alloy C-4)
02	With ball valve
03	With lip valve
04	With minimum immersion depth
06	With removable injection pipe
08	High process water temperature
Max. pressure [bar]	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
Body material	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
PP	Polypropylene
PVC	Polyvinyl chloride
PV	PVDF
SS	Stainless steel 1.4401 / 1.4571
Y	2.4610 (Alloy C-4)
Gasket material	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
E	EPDM
V	FKM
T	PTFE
Valve ball material	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
C	Ceramic
G	Glass
SS	Stainless steel 1.4401 / 1.4571
T	PTFE
Y	2.4610 (Alloy C-4)
Opening pressure	
INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
4	Size 01/02/03: 0.7 bar Size 04/05: 0.6 bar

Dosing line connection

INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
U2	Union nut G 5/8 with parts for hose connection 4/6 mm, 6/9 mm, 6/12 mm, 9/12 mm
U7	Union nut G 5/8 with parts for hose connection 0.17" x 1/4"; 1/4" x 3/8"; 3/8" x 1/2"
A	Union nut G 5/8 with threaded connection Rp 1/4, internal thread
V	Union nut G 5/8 with threaded connection 1/4 NPT, internal thread
U3	Union nut G 5/4 with parts for hose connection 19 or 20 mm or glued pipe connection 25 mm
A7	Union nut G 5/4 with threaded connection 3/4 NPT, external thread
A1	Union nut G 5/4 with threaded connection Rp 3/4, internal thread
A3	Union nut G 5/4 with threaded connection 3/4 NPT, internal thread
K	Union nut G 2 with parts for glued pipe connection 40 mm
B5	Union nut G 2 with parts for welded pipe connection 40 mm
A2	Union nut G 2 with threaded connection Rp 1 1/4 internal thread
A4	Union nut G 2 with threaded connection 1 1/4 NPT internal thread
A8	Union nut G 2 with threaded connection 1 1/4 NPT external thread
B6	4/6 mm cutting ring connection
C2	8/10 mm cutting ring connection
C9	10/12 mm cutting ring connection
X	No connections included

Process line connection

INJ.UNIT 0300-10 PVC/E/C 4U3-30/120	
10	G 1/4
11	1/4 NPT
20	G 1/2
21	1/2 NPT
30	G 1
31	1 NPT
40	G 2
41	2 NPT
70	DN 80 flange

Immersion depth [mm]

INJ.UNIT 0300-10 PVC/E/C 4U3-30/120

5. Maintaining the product

The product must only be serviced by authorised and qualified persons.

When working with chemicals, the accident prevention regulations applicable at the installation site must be applied.



Observe the chemical manufacturer's safety data sheets when handling chemicals.

When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

Observe section [3.1 Handling the product](#).

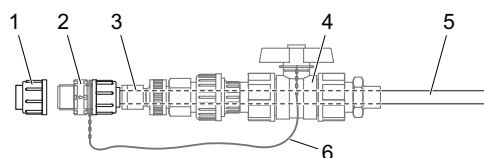
Wipe up spilled liquid immediately to avoid slipping hazard.

5.1 Maintenance schedule

Interval	Task
After 8000 operating hours or at least every 12 months or in case of faults	Clean the non-return valve. See section 5.3 Cleaning the non-return valve .

5.2 Removing and cleaning the removable injection pipe

This section applies to injection units with removable injection pipe up to 60 l/h.



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Fig. 12 Injection unit with removable injection pipe

Pos.	Description
1	Dosing line connection set
2	Spring-loaded non-return valve
3	Clamping ring connection
4	Ball valve
5	Injection pipe
6	Safety cord



When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.



In normal operation, the injection pipe passes through the ball valve and the process line connection. Do not try to close the ball valve in normal operation, otherwise the injection pipe can be damaged.

- Shut down the dosing system.
 - Make sure that the system cannot be accidentally started.
- Flush the dosing line.
- Make sure that the dosing system is pressureless.



CAUTION

Chemical hazard

Minor or moderate personal injury
- Collect escaping dosing medium safely.

- Remove the dosing line connection (1).

CAUTION

Pressurised system

Minor or moderate personal injury

- After opening the clamping ring connection (3), the injection pipe (5) can be pushed out of the injection unit by the pressure of the process water. The safety cord (6) stops the injection pipe movement to avoid that process water escapes.



- Hold the non-return valve in position and carefully open the clamping ring connection (3), until you can move the injection pipe.
 - Pull the injection pipe (5) only so far out of the injection unit, until you can close the ball valve (4).
 - Close the ball valve (4).
 - Remove the non-return valve (2) with the safety cord (6).
 - Pull out the injection pipe (5) completely.
 - Clean the injection pipe with an appropriate cleaning agent.
- Before reassembling make sure, that all parts are clean, dry and undamaged.

5.3 Cleaning the non-return valve

This section applies to injection units with a maximum flow rate of 60, 460, 1500 and 4000 l/h and high-pressure injection units.



When working on the product or connections and lines, always wear protective clothing (e.g. safety goggles and gloves). The system must be pressureless.

- Shut down the dosing system.
 - Make sure that the system cannot be accidentally started.
- Flush the dosing line.
- For injection units with removable injection pipe:
 - Pull out the injection pipe and close the ball valve. See section [5.2 Removing and cleaning the removable injection pipe](#).
- For injection units with ball valve:
 - Close the ball valve.
- For injection units without ball valve:
 - Empty the process line.
- Make sure that the dosing line is pressureless.



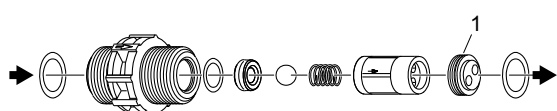
CAUTION

Chemical hazard

Minor or moderate personal injury
- Collect escaping dosing medium safely.

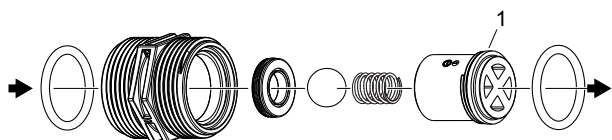
- Remove the dosing line connection.
- Remove the non-return valve.

9. For injection units up to 60, 460 and 1500 l/h and high-pressure injection units:
- Unscrew part (1), dismantle the valve and clean all parts with an appropriate cleaning agent. See figures 13, 14, 15.



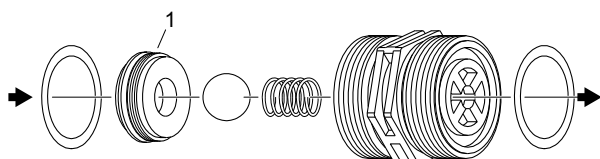
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Fig. 13 Non-return valve for injection units up to 60 l/h and high-pressure injection units



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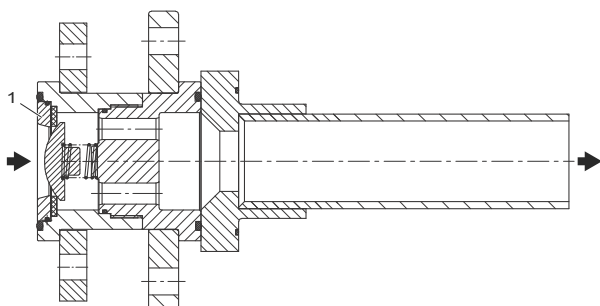
Fig. 14 Non-return valve for injection units up to 460 l/h



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Fig. 15 Non-return valve for injection units up to 1500 l/h

10. For injection units up to 4000 l/h:
- Remove the valve seat (1). See fig. 16.
 - Dismantle the valve and clean it with an appropriate cleaning agent.



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Fig. 16 Non-return valve for injection units up to 4000 l/h

Before reassembling make sure, that all parts are clean, dry and undamaged.

6. Technical data

The flow rate values apply to liquids with a viscosity similar to water.

6.1 Permissible media temperature

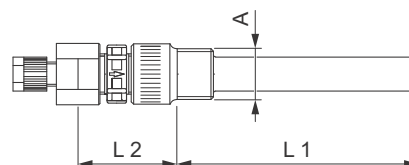
Body material	Maximum operating pressure	
	up to 10 bar	> 10 bar
PVC, PP	0 °C to +40 °C	0 °C to +20 °C
PVDF	-10 °C to +40 °C	-10 °C to +20 °C
Stainless steel	90 °C	
	145 °C (short-term with max. 2 bar)	

6.2 Storage and ambient temperatures

	Maximum operating pressure	
	up to 10 bar	> 10 bar
Min./Max. ambient temperature [°C]	0/+40	0/+20
Min./Max. storage temperature [°C]	-10/+50	

6.3 Standard injection units up to 6 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	6
Maximum operating pressure	[bar]	10
Spring material		Tantal



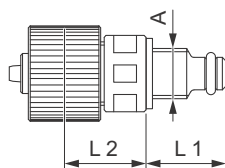
TM06 8811 1117

Fig. 17 Standard injection unit

Materials			Dimensions		
Body	Gasket	Ball	A	L 1 [mm]	L 2 [mm]
PVDF	FKM	Ceramic	G 1/2	100	41
	EPDM	Ceramic	1/2 NPT	300	
	PTFE	Ceramic			

6.4 Injection units with minimal immersion depth up to 6 l/h

Maximum flow rate	[l/h]	6
Maximum operating pressure	[bar]	10



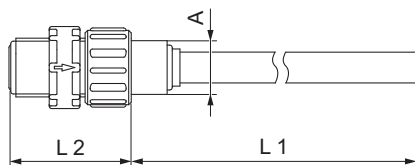
TM06 8810 1117

Fig. 18 Injection unit with minimal immersion depth

Materials			Dimensions		
Body	Gasket	O-ring	A	L 1 [mm]	L 2 [mm]
PVC	FKM	FKM	G 1/4	21	22
	EPDM	FKM			
PP	FKM	FKM	1/4 NPT	36	22
PVC	FKM	FKM			
	EPDM	FKM			
PP	FKM	FKM			

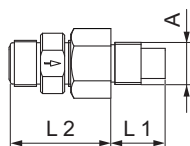
6.5 Standard injection units up to 60 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	60
Spring material		Tantal



TM06 8655 1017

Fig. 19 Standard injection unit, PP, PVC and PVDF version



TM06 8656 1017

Fig. 20 Standard injection unit, stainless steel version

Max. operating pressure [bar]	Materials			Dimensions		
	Body	Gasket	Ball	A	L 1 [mm]	L 2 [mm]
16	PP	FKM	Ceramic	G 1/2 1/2 NPT	100 300	47
		EPDM	Ceramic			
	PVC	FKM	Ceramic			
		EPDM	Ceramic			
		PTFE	Ceramic			
		FKM	Ceramic			
	PVDF	EPDM	Ceramic			
		PTFE	Ceramic			
100	SS ¹⁾	PTFE	SS ²⁾	G 1/2 1/2 NPT	27	50
	Alloy C-4 ³⁾	PTFE	Alloy C-4	G 1/2	100	47

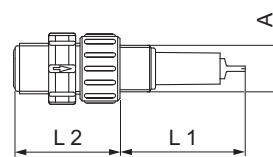
1) Stainless steel 1.4571

2) Stainless steel 1.4401

3) Injection pipe material: PVC

6.6 Injection units with lip valve up to 60 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	60
Maximum operating pressure	[bar]	16
Spring material		Tantal



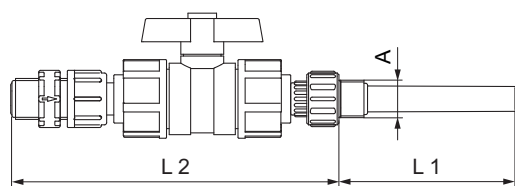
TM06 8657 1017

Fig. 21 Injection unit with lip valve

Materials				Dimensions		
Body	Gasket	Ball	Lip	A	L 1 [mm]	L 2 [mm]
PVC	FKM	Ceramic	FKM	G 1/2 1/2 NPT	~55	49

6.7 Injection units with ball valve up to 60 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	60
Spring material		Tantal



TM06 8658 1017

Fig. 22 Injection unit with ball valve

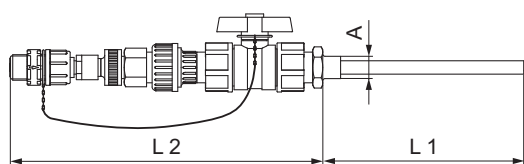
Max. operating pressure [bar]	Materials			Dimensions		
	Body	Gasket	Ball	A	L 1 [mm]	L 2 [mm]
16	PVC	FKM	Ceramic	G 1/2	100	183
		EPDM	Ceramic	1/2 NPT		
64	SS*	PTFE	SS**		27	138

* Stainless steel 1.4571 and 1.4408

** Stainless steel 1.4401

6.8 Injection units with removable injection pipe up to 60 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	60
Maximum operating pressure	[bar]	10
Spring material		Tantal



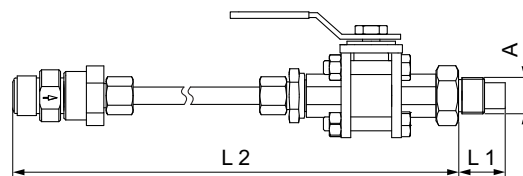
TM06 8659 1017

Fig. 23 Injection unit with removable injection pipe

Materials			Dimensions		
Body	Gasket	Ball	A	L 1 [mm]	L 2 [mm]
PVC	FKM	Ceramic	G 1/2	185	280
	EPDM	Ceramic	1/2 NPT		

6.9 Hot-injection units with ball valve up to 60 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	60
Maximum process water temperature	[°C]	120
Spring material		Tantal
Ball valve material		Stainless steel 1.4408
Injection pipe material		Stainless steel 1.4571
Cooling pipe material		Stainless steel 1.4571



TM06 8660 1017

Fig. 24 Hot-injection unit with ball valve

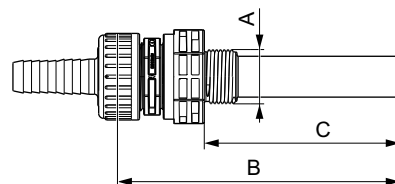
Max. operating pressure [bar]	Materials			Dimensions		
	Body	Gasket	Ball	A	L 1 [mm]	L 2 [mm]
16	PVDF	PTFE	Ceramic	G 1/2	27	1158
64	SS*	PTFE	SS**	1/2 NPT		

* Stainless steel 1.4571

** Stainless steel 1.4401

6.10 Standard injection units up to 460 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	460
Spring material		2.4610 (Alloy C-4)



TM06 8661 1017

Fig. 25 Standard injection unit

Max. operating pressure [bar]	Materials			Dimensions		
	Body	Gasket	Ball	A	B [mm]	C [mm]
10	PVC	FKM	Ceramic	G 1 1/2 NPT	173	120
		EPDM	Ceramic			
		PTFE	Ceramic			
	PP	FKM	Ceramic			
		EPDM	Ceramic			
		FKM	Ceramic			
16	PVDF	EPDM	Ceramic			
		PTFE	Ceramic			
16	SS*	PTFE	SS**			

* Stainless steel 1.4571 and 1.4408

** Stainless steel 1.4401

6.11 Injection units with ball valve up to 460 l/h

Approximate opening pressure	[bar]	0.7
Maximum flow rate	[l/h]	460
Maximum operating pressure	[bar]	10
Spring material	2.4610 (Alloy C-4)	

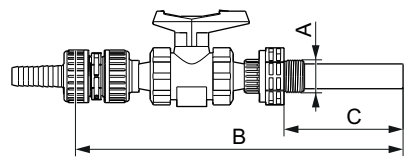


Fig. 26 Injection unit with ball valve, PVC version

TM06 8662 1217

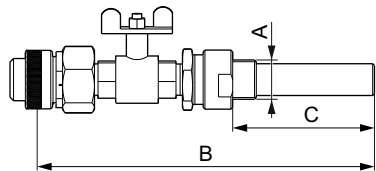


Fig. 27 Injection unit with ball valve, stainless steel version

TM06 8663 1217

Materials			Dimensions		
Body	Gasket	Ball	A	B [mm]	C [mm]
PVC	FKM	Ceramic	G 1 1 NPT	330	120
	EPDM	Ceramic			
SS*	PTFE	SS**		285.5	120

* Stainless steel 1.4571 and 1.4408

** Stainless steel 1.4401

6.12 Injection units up to 1500 l/h

Approximate opening pressure	[bar]	0.6
Maximum flow rate	[l/h]	1500
Maximum operating pressure	[bar]	10
Spring material	2.4610 (Alloy C-4)	

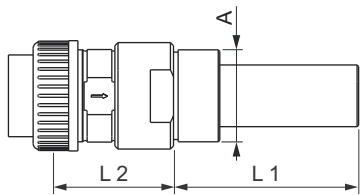


Fig. 28 Injection unit up to 1500 l/h, PP, PVC and PVDF version

TM07 0543 0218

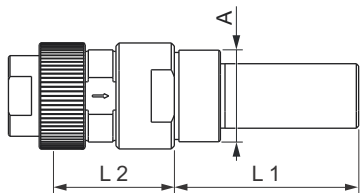


Fig. 29 Injection unit up to 1500 l/h, stainless steel version

TM06 8751 0218

Materials			Dimensions		
Body	Ball	Gasket	A	L 1 [mm]	L 2 [mm]
PVC	FKM	Glass	G 2 2 NPT	120	78
	EPDM	PTFE			
	PTFE	PTFE			
PP	FKM	Glass			
	EPDM	PTFE			
	PTFE	PTFE			
PVDF	FKM	PTFE			
	EPDM	PTFE			
	PTFE	PTFE			
Stainless steel	PTFE	Stainless steel			

6.13 Injection units up to 4000 l/h

Approximate opening pressure	[bar]	0.1
Maximum flow rate	[l/h]	4000
Maximum operating pressure	[bar]	6
Spring material	PTFE coated stainless steel 1.4310	

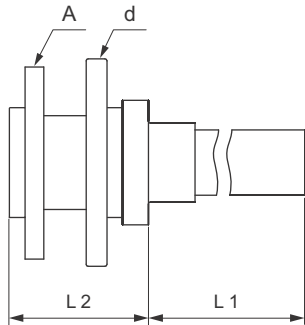


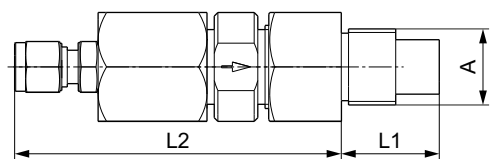
Fig. 30 Injection unit up to 4000 l/h

TM06 8752 1017

Materials				A	d	L 1 [mm]	L 2 [mm]
Body	Ball	Gasket	Seat				
PVC	PVC	FKM	PTFE	Flange DN 65	Flange DN 80	220	134
PP	PP						

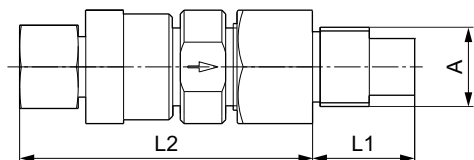
6.14 High-pressure injection units

Approximate opening pressure [bar] 0.7
Spring material 2.4610 (Alloy C-4)



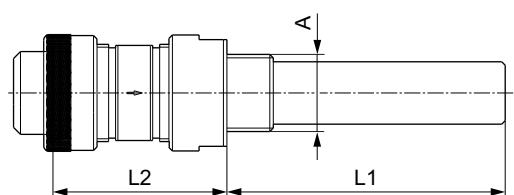
TM07 0553 0218

Fig. 31 High-pressure injection unit, flow rate 3.3 l/h



TM07 0552 0218

Fig. 32 High-pressure injection unit, flow rate 55 l/h



TM07 0551 0218

Fig. 33 High-pressure injection unit, flow rate 220 l/h

Flow rate [l/h]	Max. operating pressure [bar]	Materials		Dimensions		
		Body / ball	Gasket	A	L 1 [mm]	L 2 [mm]
3.3	200	SS*	PTFE	G 1/2	27	90
55	200	SS*	PTFE	G 1/2	27	77.5
220	100	SS*	PTFE	G 1	120	75

* Stainless steel

7. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

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